Unsupervised Human Segmentation for Sports

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Abstract

Implements the FreeSolo framework to learn class-agnostic instance segmentation from an unlabeled ultimate frisbee video. This work introduces the application of a novel spatio-temporal loss to be added to objective function of both the dense feature extractor (FreeMask) and to the instance segmenter of self-supervised SOLO. The addition of this to the objective allows for more consistent predictions across frames leading to more robust information extraction from videos. Overall, this method will allow for automatic digestion and synthesis of sport videos in a useful form.

1 Introduction

In this project, I crafted a network that can learn to segment humans and other objects in a video, without any annotations, and track different instances across the video— even when briefly occluded or out of frame. This project is motivated by the large quantities of sports video that is collected, but unlabeled, and aims to offer an analytical tool to digest the video and highlight important metrics and moments. To first test this network, I plan to tailor it to analyze ultimate frisbee film. This model could track the different players and the disc to determine when a score occurs, how many passes a given player completed, how many D's an athlete generated, etc. This network would be able to offer comprehensive analytics from videos of any sports team and could have even broader applications in other domains. It is more generally applicable as the same framework can be applied to detect players of any sport and the object (ball, bat, racquet, etc.) that is utilized in that game and then can similarly generate a transcript of important events. In this project, the scope of the project is narrowed to simply segmenting the athletes and the ultimate frisbee disc.

2 Methodology

This model uses a pre-trained backbone, specifically DenseCL, with all but the last couple of layers frozen. The model is then retrained with an additional spatio-temporal loss added to the objective. The extracted features from the backbone will then be passed to a custom model that implements the FreeSolo[5] framework. This framework allows us to learn class-agnostic instance segmentation without any annotations. This is done by using the coarse masks acquired through the FreeMask procedure as weak annotations for a semi self-supervised training. Self-Supervised SOLO trains an instance segmenter using the segmentation masks and semantic embeddings, i.e., feature embeddings with high-level semantics, from Free Mask. To this instance segmenter, we once again apply a spatio-temporal loss[2] to help enforce object tracking across frames.

Prior to passing through the pipeline, it is important to pre-process and clean the data. To do this, I scraped the Premier Ultimate League's YouTube for high quality, consistent video of ultimate frisbee. I then went through and cut the video by into individual points— each to be treated as a separate dataset— and converted the mp4 file into images to feed into the model. With the images, I followed a standard pre-processing pipeline to resize the images to be compatible with the network architecture and then I normalized the images.

3 Related Works

3.1 DenseCL

A dense self-supervised learning method that works directly at the pixel-level (or local features) by looking at the correspondence between local features. It implements self-supervised learning by optimizing a pairwise contrastive (dis)similarity loss between two views of input images[6] and is a robust feature extractor.

3.2 FreeSolo

Generates a instance bounding box labels and segmentation masks from a self-supervised pretrained model using coarse masks generated from Free Mask as a weak annotation. FreeMask is able to generate object masks directly from unlabeled images. Then train the SOLO[4]-based instance segmenter using the coarse masks and semantic embeddings from Free Mask, with novel design elements including weakly-supervised design, self-training, and semantic embedding learning.

3.3 Instance Segmentation

Solves the problem of identifying individual instances of different classes in an image. Prior works include top-down methods[3] which solve the problem from the perspective of object detection— detecting the bounding box of objects first and then segmenting. Or bottom-up methods[1] which view the task as a label-then-cluster problem— by learning per-pixel embeddings first and then clustering them into groups. Recent methods combine the two approaches for faster inferences.

3.4 Spatial-Temporal Loss

This is a novel loss function that is used in this work. It utilizes the video format to enforce small differences in pose between instances across frames. It can take in any pairwise loss function and a set of images (that are temporally connected, i.e. images 3-9) and ensures that the bounding box and pixel wise predictions across the frames are consistent. Because of the nature of video data, this is something that we can assume is very close to true as there is not that much movement within consecutive frames of a video.

4 Datasets

Before, I had planned on using a hand-collected dataset of sMITe film that has videos across games from many different tournaments and scrimmages over the course of the spring 2023 season. However, after some inital data processing, I decided to instead scrape the Premier Ultimate League's (PUL's) youtube channel for full length, professional games of higher video quality. After the inital data processing, I noticed that the quality of the video that we had collected as a team was not consistent and added more variables to the problem.

5 Objectives and Metrics

The goal of this project is to correctly do instance segmentation on a given video without any labeled data and to keep track of individual humans and discs across frames. As this project is unsupervised, we will have to evaluate the results qualitatively by drawing annotations on top of the video and determining if it passes the eye test.

First, we can see that in an image from the beginning of the video (1) that we are able to very successfully segment both the humans and the frisbee in the image. In Figures 2 and 3 we are able to see the advantages and disadvantages of adding the spatio-temporal loss to the objective function. We see that with the new loss, the predictions across the frames are a lot more consistent— however, people who were successfully segmented in other frames are no longer necessarily segmented correctly. I hypothesize that this is because the loss is pushing the network to really only choose instances that it is very confident about. So, in this trade-off, we end up losing some accuracy on some frames in exchange for a more con-

Figure 1: Human: purple, Frisbee: orange

Figure 2: FreeSolo Architecture w/out Spatio-Temporal Loss

sistency across the frames.

Another interesting result is that even though we are able to identify the frisbee in the Figure 1, we are unable to do so in any of the scenes presented later on. Similarly, in the the three frames shown in Figures 2 and 3, many people are left unsegmented. This will be discussed further in the next section.

6 Insights and Future Work

One reason for the lack of frisbee segmentation and all humans being segmented is that it is hard for the original network to pick up instances far away (in the background) or if they are in a crowd. This is consistent with the results that I produced. Further, the introduction of the new loss function requires that confidence be even higher before predicting an instance— as such, in these cases where the humans are far away or obscured or the frisbee is really small, confidence is low and as such not predicted.

Other insights are that the use of pre-trained model and fine-tuning last couple of layers makes model very computationally inexpensive. And, enforcing temporal and spatial consistency reduces number of segmentations found, but are more consistent across frames.

For future work, I would like to tune the hyperparameter controlling the scale of the queries generated by FreeMask (in the coarse mask prediction that is used to weakly annotate the data) to see if we can get better results with objects that are far away or small.

7 Conclusion

In conclusion, I was able to use the FreeMask and FreeSolo pipeline to generate instance segmentations from an unlabeled dataset of frisbee images. The architecture includes a novel loss that enforces consistency across frames. When combined with other techniques in the future, this work will allow for more robust video digestion and analysis.

Figure 3: FreeSolo Architecture with Spatio-Temporal Loss

References

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